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Department of Mining Engineering

End-Semester Examination (Winter Session) 2023-24 for B.Tech. Students (45 nos.)

Subject: Advanced Underground Mining Methods (MND 411)   Venue: NLHC LH 07

Time: 180 Minutes (04.00-07.00 PM)   Date: 05.05.2024   Maximum Marks: 96

Answer questions of all the sections

Question No.	Section A Answer any four questions [14×4=56 Marks]	Marks
1. UGC pg13 UGC PG16	**a.** Describe the CRIP method in the UCG application with the schematic diagram. **b.** Discuss the various factors affecting the process performance in UCG technology.	[4+10]
2. Highwall Mining pg pg 7 ;16; 17 chart gpt	**a.** Discuss the applicability, working methods and mechanism vis-à-vis major infrastructural needs for successful deployment of Highwall Mining Technology. **b.** Briefly enumerate the technological advancement witnessed in longwall technology at the Adriyala Longwall Project (ALP) of SCCL Mine.	[8+6]
3.	What are the different slicing methods available to mine thick coal seams? Compare the inclined slicing method in ascending to descending order concerning their applicability, mode of development and depillaring in a given mining condition.	[3+11]
4. pg-176 metal pdf pg 166 metal pdf	**a.** What is Livingston's crater theory? Why it is so important to the VCR method? **b.** Discuss the various failure modes of the pillar with correct reasoning and schematics.	[7+7]
5. pg 40 metal pdf	**a.** Provide a comparative note on Empirical Vs. Qualitative method for selection of a mining method. **b.** Select the appropriate mining method from the ore deposit data given in Table 2 using the Nicholas Approach. (Refer Table 3-6 for the application of Nicholas Ranking Method)	[6+8]
	Section B **Answer all the questions ~~[2×12 =24 Marks]~~**	
6.	**a.** A longwall face of Mine 'X' is planned to operate under 350m cover depth with a face length of 200m and a face height of 3.5m. The physico-mechanical properties of the various roof layers are given in Table 1. Estimate the span of the main fall for the given mining condition using the relation:	[15+5+4]

Boshkov Wright Hartman

~~WEB~~ UBC

Elastic 3GPa

$$E = (0.31 \times \sigma_c)\ GPa$$

. Table headings

Figure :- Overlay of GCP's on High resolution Satellite Image data (Copernicus website)

Number

Bullet

$$W_S = 324F_L^{-0.43}D^{-1.38}\sqrt{\frac{\sigma_{mIR}\times T_{IR}}{\gamma_{IR}}}\sqrt{\frac{\sigma_{mMR}\times T_{MR}}{\gamma_{MR}}}$$

Figure :- Map showing ROI of Sentinel overlay on

Figure showing flow chart of processing in SNAP

b. Following the same strata characteristics if the mine management plans to operate the next three longwall faces by changing the face length to 225m, 275m, and 300m, then what would be the span of the main fall for these new panels?

c. In the given relation, why are we not considering the variable pertaining to coal strength and mining height for the determination of the main fall span in the given geo-mining condition?

Table heading

Overall methodology of Processing of SAR Data

7\. [3+3+5+5]

a. A room and pillar operation is planned for a depth of 450m in a stratum that has a compressive strength estimated as 26.37 MPa. Entries and crosscuts are planned at 6.5m. Determine the possible extraction ratio. (Assume, γ=0.023 MPa/m)

b. If the pillars are square in size, for the estimated extraction ratio in 7(a), determine the pillar dimension.

c. Suppose the pillar safety factor of 1.5 is required for the room and pillar method with the square pillar, determine the extraction ratio and pillar dimension.

d. With the same condition (refer to conditions, 7a and 7c), suppose the pillars are three times as long as they are wide, determine the extraction ratio and pillar dimension.

Study Area Points

Govind 2023

Ground Control pts

Subsidence monitoring

Source :- Govind, 2022

Bullet pts

Figure :- Contrast Displacement Map at Jharia Region using Sentinel data

Table 1: Physico-mechanical characteristics of the roof layers

Layers	Thickness (m)	Density (Kg/m³)	RQD (%)	Compressive Strength (MPa)	Tensile Strength (MPa)
Coal	3.5	1400	40	23.16	1.54
Immediate Roof	35.23	2045	80	12.25	1.10
Main Roof	17.46	2050	76	12.89	1.13
Overburden	297.31	2089	74	13.21	1.26

Assume, G = 0.03°C/m, $\beta = 8\times10^{-6}$ / °m, υ = 0.25, γ = 0.023 MPa/m

Topography of Study Area in ~~GIS~~ up

:- Map showing the Study Area the JCF

3.2
6.5
2

1 → Objective 1 Topography Elevation Grid

Slide :- 2 → Bullet points

3 → Figure 1

Working methodology of Radar S

Mapping of Study Area

Slides Bullet points

Objectives → 1 page